The Normal Distribution

ClipSAT Printable Practice Suite (clipsat.org) • Topic 3.03

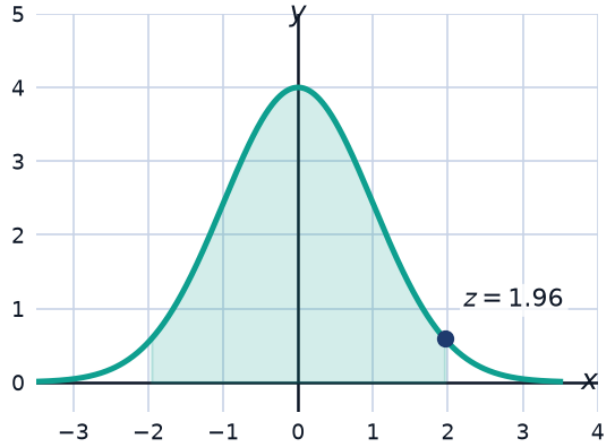
Part I — Free-Response Questions (FRQs): Foundational Fluency (1–7)

1 [Foundational] • Target: Properties of the normal distribution

State three key properties of a normal distribution's shape.

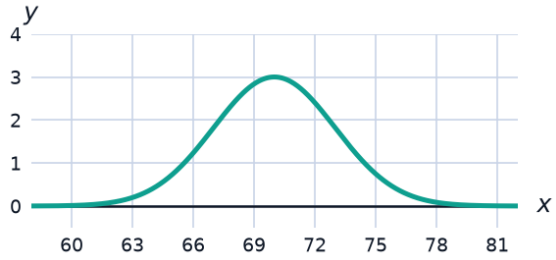
2 [Foundational] • Target: Properties of the normal distribution

Explain what the two parameters μ and σ control in a normal distribution's shape.



3 [Foundational] • Target: The 68-95-99.7 rule

Adult male heights are approximately normal with mean 70 inches and standard deviation 3 inches. Using the 68-95-99.7 rule, find the range of heights containing the middle 95% of adult men.



4 [Foundational] • Target: The 68-95-99.7 rule

Using the same distribution, find the approximate percentage of adult men taller than 76 inches.

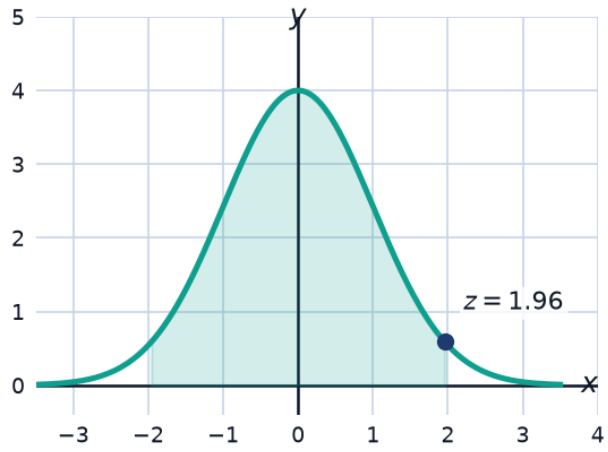
5 [Foundational] • Target: Standardizing and using the normal distribution

Using the same height distribution (mean 70, standard deviation 3), find the z-score for a height of 77.5 inches, and use a standard normal table value of $P(Z < 2.5) \approx 0.9938$ to find the proportion of men shorter than 77.5 inches.

© ClipSAT — clipsat.org | All Rights Reserved

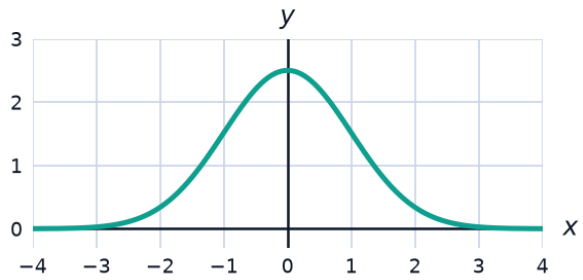
6 [Foundational] • Target: Standardizing and using the normal distribution

Find the proportion of men between 65 and 75 inches tall, given $P(Z < 1.67) \approx 0.9525$ and $P(Z < -1.67) \approx 0.0475$.



7 [Foundational] • Target: Visualizing the standard normal distribution

The graph shows the standard normal distribution ($\mu = 0$, $\sigma = 1$), with the area to the left of $z = 1$ shaded (though shading itself is not shown, imagine it visually).



a Using $P(Z < 1) \approx 0.8413$, find the area under the curve to the right of $z = 1$.

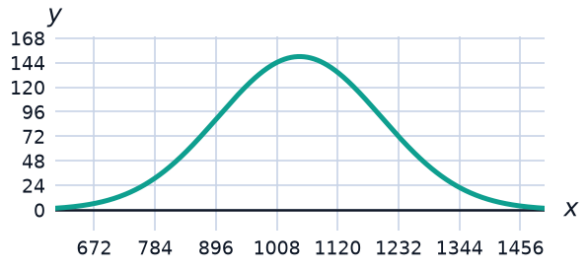
b Explain, using the curve's symmetry, why $P(Z < -1) = P(Z > 1)$.

Part II — Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14)

© ClipSAT — clipsat.org | All Rights Reserved

8 [Intermediate] • Target: Working backward: finding values from percentiles

SAT scores are approximately normal with mean 1050 and standard deviation 150. A student scores at the 90th percentile, corresponding to $z \approx 1.28$. Find that student's approximate score.

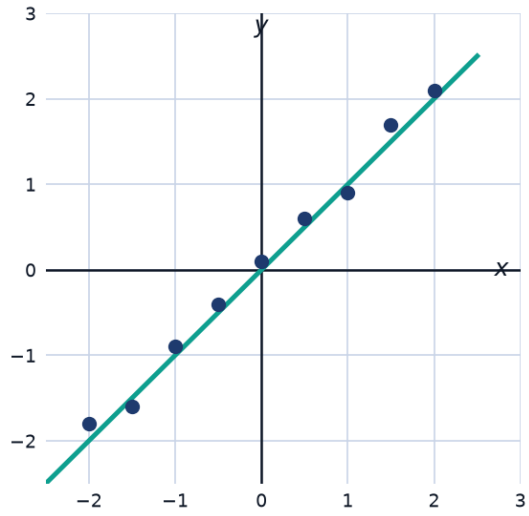


9 [Intermediate] • Target: Working backward: finding values from percentiles

Using the same distribution, find the score that separates the bottom 10% of scores, given that $z \approx -1.28$ corresponds to the 10th percentile.

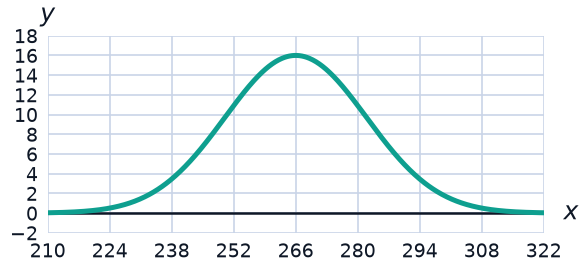
10 [Intermediate] • Target: Assessing normality

Explain how a normal probability plot (or normal quantile plot) can be used to assess whether a data set is approximately normal, and what a roughly straight-line pattern indicates.



11 [Intermediate] • Target: Applying the normal distribution to real data

Pregnancy lengths are approximately normal with mean $\mu = 266$ days and standard deviation $\sigma = 16$ days. Find the z-score for a pregnancy lasting 250 days.



© ClipSAT — clipsat.org | All Rights Reserved

12 [Intermediate] • Target: Applying the normal distribution to real data

Using the same pregnancy-length distribution and the 68-95-99.7 rule, find the approximate percentage of pregnancies lasting between 234 and 298 days.

13 [Intermediate] • Target: Combining two independent normal random variables

A vending machine dispenses a soda whose volume is normally distributed with mean 355 mL and standard deviation 4 mL, independently of a snack bag whose weight is normally distributed with mean 50 g and standard deviation 2 g. If T represents the combined "total" formed by adding the soda volume (in mL) and snack weight (in g) as a single number, find the mean of T. (This is a simplified illustrative combination, not a real physical unit.)

14 [Intermediate] • Target: Combining two independent normal random variables

Battery lifespans are normally distributed with mean 500 hours and standard deviation 40 hours. Find the z-score for a battery lasting 460 hours, and describe its relative position.

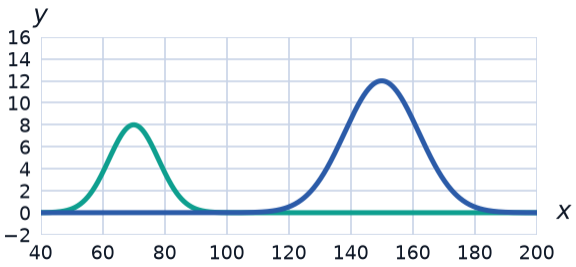
Part III — Free-Response Questions (FRQs): Challenging & Cross-Lesson Synthesis (15–20)

15 [Challenging] • Target: Combining two independent normal random variables • Cross-Lesson: 1.02 Z-Scores & Linear Transformations

A company manufactures resistors with resistance values approximately normal, mean 100 ohms and standard deviation 5 ohms. A resistor is considered defective if its resistance is below 90 ohms or above 110 ohms. Using the 68-95-99.7 rule, estimate the percentage of resistors that are defective, and explain your reasoning step by step.

16 [Challenging] • Target: Combining two independent normal random variables • Cross-Lesson: 1.03 Describing Distributions & Linear Regression

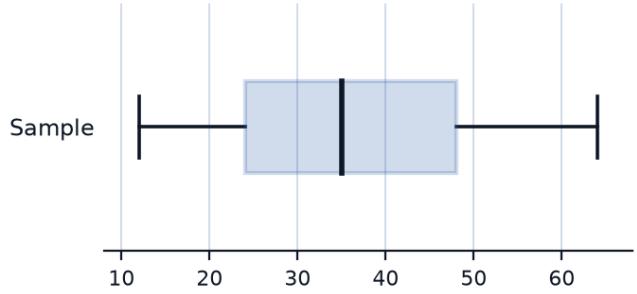
Two students are comparing their standardized test scores from different tests. Student X scored 78 on a test with mean 70 and standard deviation 8. Student Y scored 165 on a different test with mean 150 and standard deviation 12. Using z-scores, determine who performed relatively better compared to their own test's distribution of test-takers, showing full reasoning.



© ClipSAT — clipsat.org | All Rights Reserved

17 **[Challenging]** • **Target: Working backward: finding values from percentiles** • **Cross-Lesson: 2.01 Sampling Methods & Bias**

A sample dataset has $Q_1 = 48$ and $Q_3 = 76$. Using the 1.5 IQR rule: (a) Determine the exact lower and upper outlier fences. (b) If the dataset includes values of 4 and 122, determine whether each is an outlier.



18 **[Challenging]** • **Target: Assessing normality** • **Cross-Lesson: 2.02 Experimental Design & Scope of Inference**

A researcher wants to investigate the relationship between the type of pet owned (Dog, Cat, Fish) and the preferred music genre (Rock, Pop, Classical). They collect data and present it in a two-way table. If the table has 3 rows and 3 columns, what is the total number of cells in the table that represent a combination of pet ownership and music genre?

19 **[Challenging]** • **Target: Applying the normal distribution to real data** • **Cross-Lesson: 3.01 Probability Rules & Conditional Probability**

A bag contains marbles of different colors. The probability of drawing a red marble is $P(\text{Red}) = \frac{2}{5}$ and the probability of drawing a blue marble is $P(\text{Blue}) = \frac{1}{3}$. If the draws are independent, what is the probability of drawing a red marble followed by a blue marble?

20 **[Challenging]** • **Target: Combining two independent normal random variables** • **Cross-Lesson: 3.02 Discrete Random Variables & Binomial Distributions**

A manufacturing company is testing a new quality control process. They have collected data on the number of defects found in batches of a product, categorized by the production shift. The data is summarized in the two-way table below: | Shift | Defects Found | No Defects Found | Total | |---|---|---|---| | Morning | 15 | 85 | 100 | | Afternoon | 25 | 75 | 100 | | Evening | 10 | 90 | 100 | | Total | 50 | 250 | 300 | If a test of independence is to be performed on this data, what is the expected number of defects found in the afternoon shift?

Part IV — Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)

© ClipSAT — clipsat.org | All Rights Reserved

21 [Foundational] • Target: Properties of the normal distribution

A normal distribution has mean 64 and standard deviation 9. Using the 68-95-99.7 rule, find the range containing the middle 95% of values.

a. $(46, 82)$

b. $(37, 91)$

c. $(46, 81)$

d. $(47, 82)$

22 [Foundational] • Target: The 68-95-99.7 rule

A normal distribution has mean 332 and standard deviation 16. Using the 68-95-99.7 rule, find the range containing the middle 99.7% of values.

a. $(268, 396)$

b. $(284, 379)$

c. $(285, 380)$

d. $(284, 380)$

23 [Foundational] • Target: Standardizing and using the normal distribution

A normal distribution has mean 457 and standard deviation 11. Using the 68-95-99.7 rule, find the range containing the middle 99.7% of values.

a. $(425, 490)$

b. $(424, 490)$

c. $(424, 489)$

d. $(413, 501)$

24 [Foundational] • Target: Visualizing the standard normal distribution

A normal distribution has mean 50 and standard deviation 14. Using the 68-95-99.7 rule, find the range containing the middle 68% of values.

a. $(22, 78)$

b. $(36, 64)$

c. $(36, 63)$

d. $(37, 64)$

25 [Foundational] • Target: Working backward: finding values from percentiles

A normal distribution has mean 138 and standard deviation 40. Using the 68-95-99.7 rule, find the range containing the middle 95% of values.

a. $(58, 218)$

b. $(18, 258)$

c. $(58, 217)$

d. $(59, 218)$

© ClipSAT — clipsat.org | All Rights Reserved

26 [Foundational] • Target: Assessing normality

A normal distribution has mean 88 and standard deviation 18. Using the 68-95-99.7 rule, find the range containing the middle 68% of values.

a. $(70, 106)$

b. $(70, 105)$

c. $(52, 124)$

d. $(71, 106)$

27 [Foundational] • Target: Applying the normal distribution to real data

A normal distribution has mean 361 and standard deviation 35. Using the 68-95-99.7 rule, find the range containing the middle 95% of values.

a. $(291, 430)$

b. $(256, 466)$

c. $(292, 431)$

d. $(291, 431)$

28 [Foundational] • Target: Combining two independent normal random variables

A normal distribution has mean $\mu = 101$ and standard deviation $\sigma = 4$. Find the approximate percentage of values below $x = 103$, to two decimal places.

a. 69.15%

b. 72.15%

c. 66.15%

d. 30.85%

Part V — Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37)

29 [Intermediate] • Target: The Empirical Rule (68-95-99.7)

A normal distribution has mean $\mu = 64$ and standard deviation $\sigma = 10$. Find the approximate percentage of values below $x = 39$, to two decimal places.

a. 0.62%

b. 2.62%

c. 3.62%

d. 99.38%

© ClipSAT — clipsat.org | All Rights Reserved

30 [Intermediate] • Target: Finding Normal Probabilities

A normal distribution has mean $\mu = 477$ and standard deviation $\sigma = 20$. Find the approximate percentage of values below $x = 527$, to two decimal places.

a. 96.38%

b. 0.62%

c. 97.38%

d. 99.38%

31 [Intermediate] • Target: Inverse Normal Calculations (Finding Values from Probabilities)

A normal distribution has mean $\mu = 174$ and standard deviation $\sigma = 16$. Find the approximate percentage of values below $x = 198$, to two decimal places.

a. 6.68%

b. 90.32%

c. 93.32%

d. 96.32%

32 [Intermediate] • Target: Assessing Normality with Normal Probability Plots

A normal distribution has mean $\mu = 338$ and standard deviation $\sigma = 20$. Find the approximate percentage of values below $x = 388$, to two decimal places.

a. 99.38%

b. 96.38%

c. 0.62%

d. 97.38%

33 [Intermediate] • Target: Properties of the normal distribution

A normal distribution has mean $\mu = 99$ and standard deviation $\sigma = 2$. Find the approximate percentage of values below $x = 104$, to two decimal places.

a. 99.38%

b. 97.38%

c. 96.38%

d. 0.62%

34 [Intermediate] • Target: The 68-95-99.7 rule

A normal distribution has mean $\mu = 343$ and standard deviation $\sigma = 4$. Find the approximate percentage of values below $x = 335$, to two decimal places.

a. 5.28%

b. 2.28%

c. 4.28%

d. 97.72%

© ClipSAT — clipsat.org | All Rights Reserved

35 [Intermediate] • Target: Standardizing and using the normal distribution

A normal distribution has mean $\mu = 237$ and standard deviation $\sigma = 2$. Find the approximate percentage of values between $x_1 = 233$ and $x_2 = 240$, to two decimal places.

a. 91.04%

b. 87.04%

c. 95.04%

d. 93.32%

36 [Intermediate] • Target: Visualizing the standard normal distribution

A normal distribution has mean $\mu = 123$ and standard deviation $\sigma = 10$. Find the approximate percentage of values between $x_1 = 133$ and $x_2 = 148$, to two decimal places.

a. 11.24%

b. 19.24%

c. 99.38%

d. 15.24%

37 [Intermediate] • Target: Working backward: finding values from percentiles

Scores on an exam are approximately normal with mean $\mu = 837$ and standard deviation $\sigma = 50$. A student scores at the 95th percentile, corresponding to $z \approx 1.645$. Find that student's approximate score.

a. 909

b. 755

c. 919

d. 929

Part VI — Multiple-Choice Questions (MCQs): Challenging & Cross-Lesson Synthesis (38–45)

38 [Challenging] • Target: Assessing normality • Cross-Lesson: 2.02 Experimental Design & Scope of Inference

Scores on an exam are approximately normal with mean $\mu = 295$ and standard deviation $\sigma = 20$. A student scores at the 10th percentile, corresponding to $z \approx -1.28$. Find that student's approximate score.

a. 279

b. 259

c. 321

d. 269

© ClipSAT — clipsat.org | All Rights Reserved

39 [Challenging] • Target: Applying the normal distribution to real data • Cross-Lesson: 3.01 Probability Rules & Conditional Probability

A company manufactures parts with a measurement approximately normal, mean 224 and standard deviation 27. A part is considered defective if its measurement is below 170 or above 278 (i.e. outside $\mu \pm 2\sigma$). Using the 68-95-99.7 rule, estimate the percentage of parts that are defective.

a. 32%

b. 2.5%

c. 5%

d. 0.3%

40 [Challenging] • Target: Combining two independent normal random variables • Cross-Lesson: 3.02 Discrete Random Variables & Binomial Distributions

A company manufactures parts with a measurement approximately normal, mean 84 and standard deviation 15. A part is considered defective if its measurement is below 39 or above 129 (i.e. outside $\mu \pm 3\sigma$). Using the 68-95-99.7 rule, estimate the percentage of parts that are defective.

a. 0.3%

b. 5%

c. 0.15%

d. 32%

41 [Challenging] • Target: The Empirical Rule (68-95-99.7) • Cross-Lesson: 3.04 Sampling Distributions & Central Limit Theorem

In a normal distribution with mean $\mu = 100$ and $\sigma = 15$, approximately what percentage of scores fall between 70 and 130?

a. 95%

b. 68%

c. 99.7%

d. 50%

42 [Challenging] • Target: The Empirical Rule (68-95-99.7) • Cross-Lesson: 4.01 Confidence Intervals & Hypothesis Testing

If heights of adult men are $N(70, 3)$ inches, what percentage are taller than 76 inches?

a. 2.5%

b. 5%

c. 16%

d. 0.15%

© ClipSAT — clipsat.org | All Rights Reserved

43 [Challenging] • Target: Finding Normal Probabilities • Cross-Lesson: 4.02 Chi-Square Tests & Regression Inference

A machine fills cereal boxes normally with $\mu = 16.2$ oz and $\sigma = 0.1$ oz. What proportion of boxes contain less than the stated label of 16.0 oz?

a. $P(Z < -2.0) \approx 0.0228$

b. $P(Z < -1.0) \approx 0.1587$

c. $P(Z > 2.0) \approx 0.9772$

d. 0.5000

44 [Challenging] • Target: Inverse Normal Calculations (Finding Values from Probabilities) • Cross-Lesson: 1.01 Measures of Center & Spread

In a normal distribution, what are the z-scores corresponding to the first quartile (Q_1) and third quartile (Q_3)?

a. $z \approx -0.67$ and $z \approx +0.67$

b. $z \approx -1.00$ and $z \approx +1.00$

c. $z \approx -0.50$ and $z \approx +0.50$

d. $z \approx -1.96$ and $z \approx +1.96$

45 [Challenging] • Target: Assessing Normality with Normal Probability Plots • Cross-Lesson: 1.02 Z-Scores & Linear Transformations

What appearance in a Normal Probability (Q-Q) Plot indicates that the sample data comes from a roughly normal population?

a. The plotted points follow a roughly linear pattern

b. The plotted points form a bell-shaped curve

c. The plotted points are completely horizontal

d. The plotted points form a curved parabolic shape

© ClipSAT — clipsat.org | All Rights Reserved